



PMI / PMBOK® 5TH EDITION STANDARD

AL-RASHEED SMART UNIVERSITY

TECHNICAL PROJECT MANAGEMENT DOSSIER

Smart Campus Portal

Centralized University Services, Single Sign-On (SSO), Automated Administrative Workflows, and Mobile-First Accessibility.

AUTHOR & PROJECT MANAGER: Nooraldeen Al-Shmasi

COURSE INSTRUCTOR & SPONSOR: Dr. Senan A. Ghallab

ACADEMIC PROGRAM: B.Sc. Information Technology (IT)
Faculty of Engineering & Technology

CURRICULAR FRAMEWORK: Information Technology Project Management (ITPM)
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2 Initiation Phase

2.1 Problem Statement

University ecosystems are plagued by a fundamental fragmentation of IT services. Current infrastructures are built on separate, disconnected systems for student registration (SIS), e-learning (LMS), library access and communication. Students and faculty must juggle multiple credentials, deal with inconsistent user interfaces, and manually transfer data between platforms.

Critical gaps addressed by this project include:

- **Authentication Friction:** Users managing up to 5 different accounts, leading to high password reset requests and IT support overhead.
- **Data Silos:** Lack of API integration between the LMS and SIS results in delayed grade synchronization and registration bottlenecks.
- **Poor Mobile Accessibility:** Legacy systems are not responsive, frustrating students who primarily access services via smartphones.

2.2 Business Case

Strategic Value Proposition The Smart Campus Portal is a single aggregation layer that consolidates all university core services in a single interface that is mobile responsive and SSO enabled. It provides quantifiable IT support cost savings, automated data flows and increased user satisfaction.

Cost Justification & Investment Analysis

A standard financial analysis was performed with a 3-year horizon and discount rate (i) of 8% to justify the institutional investment in this portal versus maintaining legacy fragmented systems (~\$20,000 per year in IT helpdesk overhead and licensing).

YEAR	COSTS (OPEX/CAPEX)	BENEFITS (SAVINGS)	NET CASH FLOW
Year 0 (Dev)	\$15,000	\$0	-\$15,000
Year 1	\$3,000	\$20,000	\$17,000
Year 2	\$3,000	\$20,000	\$17,000
Year 3	\$3,000	\$20,000	\$17,000

Net Present Value (NPV):

Formula: $NPV = \sum R_t / (1+i)^t$

$NPV = -\$15,000 + (\$17,000/1.08) + (\$17,000/1.166) + (\$17,000/1.260)$

$NPV = -\$15,000 + \$15,740 + \$14,579 + \$13,492 = \$28,811.$

Decision Rule: Since $NPV > 0$, the project is financially acceptable.

Return on Investment (ROI):

Formula: $ROI = (Total\ Benefits - Total\ Costs) / Total\ Costs$

Total Costs = \$24,000. Total Benefits = \$60,000.

ROI = $(\$60,000 - \$24,000) / \$24,000 = 150.0\%$ over 3 years.

Payback Period:

The initial investment of \$15,000 is fully recovered within 11 months of Year 1 ($\$17,000/12 = \$1,416/\text{mo}$).

2.3 SMART Objectives

ID	OBJECTIVE	S	M	A	R
OBJ-1	Integrate SIS, LMS, and Library via SSO	Y	3 Sys	Y	Y
OBJ-2	Achieve sub-2-second portal load times	Y	<2s	Y	Y
OBJ-3	Reduce IT password reset tickets by 60%	Y	60%	Y	Y
OBJ-4	Ensure 99.9% uptime during registration week	Y	99.9%	Y	Y

2.4 Project Charter

Attribute	Description
Project Name	Smart Campus Portal
Project Manager	Nooraldeen Al-Shmasi
Project Sponsor	Dr. Senan A. Ghallab
Total Duration	16 Weeks (Academic Schedule)
Methodology	Iterative SDLC / Agile Prototyping
In-Scope	SSO integration, Web Portal UI/UX, SIS/LMS API mapping, Mobile-responsive frontend, Notifications
Out-of-Scope	Replacing backend legacy systems, native iOS/Android mobile apps
Constraints	Fixed 16-week timeline, limited academic budget

3 Stakeholder Management

3.1 Stakeholder Register

ID	STAKEHOLDER	ROLE	INTEREST	INFLUENCE
SH-01	Dr. Senan Ghallab	Sponsor	Very High	Very High
SH-02	Nooraldeen Al-Shmasi	PM / Lead Dev	Very High	Very High
SH-03	University IT Dept	Infrastructure Support	High	High
SH-04	Students & Faculty	End Users	Very High	Medium
SH-05	System Vendors (LMS)	External API Providers	Low	Medium

3.2 Power / Interest Grid

KEEP SATISFIED (High Power, Low Interest) SH-05: System Vendors	MANAGE CLOSELY (High Power, High Interest) SH-01: Dr. Senan Ghallab SH-02: Nooraldeen Al-Shmasi SH-03: IT Dept
MONITOR (Low Power, Low Interest) None in this phase.	KEEP INFORMED (Low Power, High Interest) SH-04: Students & Faculty

3.3 Cooperation-Threat Matrix

Evaluates stakeholders based on potential for cooperation vs threat.

SUPPORTIVE (High Coop, Low Threat) SH-01, SH-02, SH-04	MIXED BLESSING (High Coop, High Threat) SH-03: IT Dept (Security policy conflicts)
MARGINAL (Low Coop, Low Threat) SH-05: External Vendors	NON-SUPPORTIVE None identified.

4 Planning Phase (Core)

4.1 Scope, WBS & RTM

Work Breakdown Structure (WBS) - 100% Rule

WBS ACTIVITY	LEVEL
1.0 Smart Campus Portal	0
1.1 Phase 1: Requirements & Design	1
1.1.1 Stakeholder elicitation	2
1.1.2 UI/UX Wireframing	2
1.1.3 API Contracts (SIS, LMS)	2
1.2 Phase 2: Core Infrastructure	1
1.2.1 SSO OAuth2 Setup	2
1.2.2 Database Schema	2
1.3 Phase 3: Development	1
1.3.1 Frontend Web Portal	2
1.3.2 Backend API Gateway	2
1.4 Phase 4: Testing & Handover	1
1.4.1 Security Audit	2
1.4.2 Usability Testing	2
1.4.3 Final Documentation	2

Requirements Traceability Matrix (RTM)

REQ ID	DESCRIPTION	WBS DELIVERABLE	TEST (AT)
REQ-01	SSO Login via University ID	1.2.1 SSO OAuth2 Setup	AT-01
REQ-02	Fetch Grades from SIS	1.3.2 Backend API Gateway	AT-02
REQ-03	Sub 2-second load time	1.3.1 Frontend Web Portal	AT-03

4.2 Schedule Management & Gantt Chart

The project spans 16 weeks utilizing fast-tracking with staggered start-to-start (SS) dependencies.

ID	TASK	DUR.	PREDECESSOR	WEEKS
T1	Requirements & Design	3W	-	W1 - W3
T2	SSO Infrastructure	3W	T1 (SS+1W)	W2 - W4
T3	Frontend UI	5W	T1	W4 - W8
T4	Backend API Mapping	5W	T2	W5 - W9
T5	System Integration	4W	T3, T4	W10 - W13
T6	UAT & Handover	3W	T5	W14 - W16

Critical Path: T1 -> T3 -> T5 -> T6 (15 Weeks).

4.3 Cost & Budget Management

CATEGORY / ITEM	EST. COST (USD)
Server Infrastructure (Cloud)	\$1,500
API Gateway Licensing	\$500
PM/Dev Resources (In-Kind Academic)	\$0
Domain & SSL	\$100
TOTAL ESTIMATED OUTLAY	\$2,100
Contingency Reserve (15%)	\$315
TOTAL PROJECT BUDGET	\$2,415

4.4 Human Resources & RACI Matrix

DELIVERABLE	PM (Nooraldeen)	SPONSOR (Dr. Senan)	IT DEPT	STUDENTS
Architecture Design	R/A	C	C	I
SSO Integration	R/A	I	C	I
Frontend Dev	R/A	I	I	I
UAT Testing	A	C	I	R
Project Sign-off	R	A	I	I

4.5 Risk Management & Risk Register

Risk Exposure = Probability (1-5) × Impact (1-5)

ID	Risk Description	Category	P	I	Score	Mitigation Strategy	Owner
RSK-01	Legacy SIS API lacks documentation	Technical	4	4	16	Early PoC testing with IT infrastructure team	Lead Developer
RSK-02	SSO token exposure vulnerability	Security	2	5	10	Enforce JWT encryption and HTTPS-only traffic	Security Lead
RSK-03	Scope creep from department heads	Management	3	3	9	Strict enforcement of the Change Control Board (CCB)	Project Manager
RSK-04	Cloud server outage during launch	Operational	2	4	8	Implement multi-AZ (Availability Zone) deployment	IT Dept
RSK-05	Delayed grade synchronization with LMS	Integration	3	4	12	Implement backend cron jobs with automated retry logic	Backend Dev

4.6 Quality Management Plan

QM-ID	Quality Metric	Target / Threshold	Verification Method	Mapped AT
QM-01	API Response Latency	< 2 seconds under normal load	Automated load testing tools on backend routes	AT-03
QM-02	Security Posture	Zero critical vulnerabilities	Automated penetration scan and manual code review	AT-04
QM-03	System Uptime	99.9% uptime during registration	Server monitoring software (e.g., Uptime Kuma)	AT-05
QM-04	UI Responsiveness	100% rendering without horizontal scroll on mobile	Device emulation testing across iOS and Android	AT-06
QM-05	SSO Integration	Successful token exchange across 3 systems	End-to-end user login flow tests	AT-01

4.7 Communications Management Plan

Audience	Communication Type	Frequency	Channel / Method	Owner
Dr. Senan Ghallab (Sponsor)	Project Status & Milestone Review	Weekly	Email report & face-to-face meeting	Project Manager
IT Department	Technical & Security Alignment	Bi-weekly	Technical briefing session	Lead Developer
Students & Faculty	Usability Feedback (UAT)	Once (Week 14)	Digital survey embedded in beta portal	Project Manager
Change Control Board	Change Request (CR) Approvals	As needed	Formal CR document submission	Project Manager

5 Monitoring & Controlling Phase

5.1 Performance Tracking KPIs (EVM)

Project performance is continuously monitored using Earned Value Management (EVM).

KPI	FORMULA	TARGET
Schedule Variance (SV)	$SV = EV - PV$	$SV \geq 0$
Schedule Performance Index (SPI)	$SPI = EV / PV$	$SPI \geq 1.0$
Cost Performance Index (CPI)	$CPI = EV / AC$	$CPI \geq 1.0$
Estimate at Completion (EAC)	BAC / CPI	$EAC \leq \text{Budget}$

5.2 Change Request Process

1. Identify Need & Submit CR Form -> 2. PM Performs Impact Analysis (Scope/Time/Cost) -> 3. CCB (Dr. Senan) Reviews -> 4. Approve/Reject -> 5. Update Baselines -> 6. Communicate.

6 Closing Phase

6.1 Acceptance Criteria Checklist

AC-ID	Acceptance Criterion	Mapped Objective	Status
AC-01	SSO authentication successfully exchanges credentials between the portal, SIS, LMS, and Library.	OBJ-1	Pending
AC-02	The portal dashboard fully loads and retrieves initial SIS user data in under 2.0 seconds.	OBJ-2	Pending
AC-03	Frontend layout utilizes responsive design to render natively on both iOS and Android devices.	OBJ-1	Pending
AC-04	Security audit and penetration test complete with zero critical or high-severity flaws.	Quality Metric	Pending
AC-05	System logs demonstrate simulated IT password reset requests are reduced by at least 60%.	OBJ-3	Pending

6.2 Lessons Learned Framework

CATEGORY	LESSON / OBSERVATION
Technical	Legacy API limitations required custom middleware mapping, delaying T4 by 1 week.
Stakeholder	Early pilot testing in W12 uncovered critical UI flaws that were easily fixed.
Risk	Early PoC (RSK-01 mitigation) successfully prevented complete architecture redesign.

6.3 Project Sign-Off

This document serves as the formal sign-off record. By signing below, the parties confirm that all project deliverables have been reviewed, acceptance criteria have been verified, and the project is formally closed.

Project Sponsor / Instructor: _____

Dr. Senan A. Ghallab

Date: _____

Project Manager / Developer: _____

Nooraldeen Al-Shmasi

Date: _____